CS17600: Data Engineering in Python

Summer 2026 syllabus

Course description

The course introduces students to programming fundamentals in Python, including loops, functions and different data types, and provides an introduction to data engineering including working with common data formats and learning the basics of data wrangling. Students will format, extract, clean, filter, transform, search, combine, summarize, aggregate, and visualize a diverse range of data sets. Python libraries including matplotlib and Pandas are used.

- **Modality:** asynchronous online. There are no required lecture or lab meeting times.
- **Credit hours:** the University expects students to invest 12-18 hours per week in preparation, in addition to time spent in class, for a three-credit hour course (in an 8-week term).
- **Prerequisites:** none.

Learning outcomes

1. Write Python code using loops, decision statements, and functions.
2. Explain how arguments are passed in Python functions and how the scope of variables impacts execution.
3. Use the operations on lists, tuples, and dictionaries to perform appropriate data manipulations.
4. Using Matplotlib, create informative plots and other data visualizations. Explain the key qualities of good visualizations.
5. Creating and manipulating DataFrames using Pandas.
6. Create Python code as well as methods in Pandas to select, search, change, and summarize data in tables.
7. Explain how to identify and fill in missing values in data.
8. Apply Pandas functions combine and merge datasets, perform a range of data aggregations, groupings and cross tabulations.
9. Given multiple data sets, demonstrate how to summarize, transform, combine the data sets, and aggregate and visualize the resulting data set.

Instructors

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Teaching assistants

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Getting help

In this class we use the “Ed” platform for both class discussion and programming assignments. You can join Ed through the *Start Here* content module on Brightspace. Ed is the recommended communication method for asynchronous help with programming assignments or general questions about the course schedule, content, or your grade. You may post anonymously on Ed, but please note that course staff members can always determine the identity of anonymous posters. You may also make your post private (visible only to course staff members). This is a **requirement** if your post includes your work for a programming assignment or is regarding your grade.

The course staff can be reached via the following email address: cs176@cs.purdue.edu. In almost all cases it is preferable for you to post (privately) on Ed instead. **Never email any TAs directly.**

Please give the course staff 24 hours to respond to emails and Ed posts during the week, and 36 hours during the weekend.

Office hours are for direct, synchronous assistance. The office hours schedule is posted in the *Start Here* content module on Brightspace. You are welcome to attend the office hours of any TA. If you learned something during office hours that might help other students, please post it to Ed for the benefit of all.

Learning resources

Readings and additional materials will be posted on Brightspace to supplement the lecture content.

- How to Think Like a Computer Scientist (TLCS): 3rd edition PDF or interactive Runestone edition
- How to Think like a Data Scientist: Runestone Academy
- Python tutorial, Pandas documentation, matplotlib pyplot tutorial
- pythontutor.com to understand what happens as the computer runs each line of code
- Data Wrangling with Python: Katharine Jarmul and Jacqueline Kazil, O'Reilly Media, 2016
- Computational and Inferential Thinking: The Foundations of Data Science: 2nd edition

Brightspace

All relevant class information, updates, and announcements will be available on the course Brightspace site.

Grading

Labs (12 × 5 points each) and homework (7 × 10 points each)

Programming assignments are meant to serve as an opportunity to practice the implementation of the concepts you learn in lecture. You will be scored according to your code implementing a general solution to the given problem as well as providing sufficient documentation in the form of comments. This score will be determined through a combination of automatic and manual grading. You will receive automatic feedback the first 20 times you submit each programming assignment. All programming assignments are completed on Ed. **Please note: we cannot accept programming assignment submissions outside of the designated submission areas that have been provided for you.**

Weekly quizzes (7 × 5 points each)

Weekly quizzes are meant to be a checkpoint of your understanding of the current lecture and programming assignment content. You will be scored according to your correct responses. All weekly quizzes are completed on Brightspace.

Surveys (8 × 1–2 points each)

Surveys are meant to gather your thoughts on the course as the semester progresses. You will be scored according to your completion in answering the questions. All surveys are completed on Brightspace.

Exams

Midterm Exam

- 120 points possible
- Format: writing code
- Content: Weeks 1 – 4 and HW 4
- Date: Wednesday July 15
- Time limit: 120 minutes

Final Exam

- 150 points possible
- Format: multiple-choice
- Content: comprehensive
- Date: Wednesday August 5
- Time limit: 120 minutes

All exams are completed on Brightspace and require the use of Respondus LockDown Browser and Monitor. The proctoring requirements are not optional and limit access to electronic resources and record both your screen and your testing environment. A camera, microphone, and high-speed internet connection are necessary.

Points

- **Labs:** 12 × 5 points = 60 points
- **Homework:** 7 × 10 points = 70 points
- **Weekly Quizzes:** 7 × 5 points = 35 points
- **Surveys:** 7 × 2 points + 1 × 1 point = 15 points
- **Midterm:** = 120 points
- **Final Exam:** = 150 points
- **Total:** = 450 points

Grading scale

- A+ ≥ 436 total points
- A ≥ 418
- A– ≥ 405
- B+ ≥ 391
- B ≥ 373
- B– ≥ 360
- C+ ≥ 346
- C ≥ 328
- C– ≥ 315
- D ≥ 270
- F < 270

The instructors reserve the right to lower the minimum score required for each letter grade. If any change is made it will not be announced until after the final exam. At no time during the semester will it be speculated if this will be done or how much any given cutoff will be lowered. You should have no expectation that all cutoffs if moved will be moved by an equal amount.

Course schedule

All assignments are due at 11:00 PM on the dates listed below and can be submitted late without penalty until 11:59 PM. There are no assignments due on Tuesdays or Thursdays.

New programming assignments become available every Monday (lab due the following Sunday, HW due the following Monday) and Thursday (lab due the following Wednesday). Exams become available at 12:01 AM on the dates listed below.

Week	Topics	Date	Mon	Wed	Fri	Sat	Sun
1	• Course overview • Introduction to computers • Variables & expressions • `print()` function • `input()` function • String slicing • `range()` function • `for` loop	June 15		Lab 1	WQ 1 Survey 1		Lab 2
2	• Conditions • Lists • More about loops	June 22	HW 1	Lab 3	WQ 2 Survey 2		Lab 4
3	• Functions • More about strings • File I/O	June 29	HW 2	Lab 5	WQ 3 Survey 3		Lab 6
4	• Sets & tuples • Dictionaries	July 6	HW 3		WQ 4 Survey 4		Lab 7
5	• Data engineering • Pandas essentials	July 13	HW 4	**Midterm**	WQ 5 Survey 5		Lab 8
6	• Cleaning data • Handling missing data • Transforming data • Merging data	July 20	HW 5	Lab 9	WQ 6 Survey 6		Lab 10
7	• Visualizations • Multi-Index	July 27	HW 6	Lab 11	WQ 7 Survey 7	**Lab 12** **HW 7**	
8		August 3		**Final** **Survey 8**			

Absences and late work

- No late work is accepted except for the reasons of documented and serious hardship.
- Do not expect assignment deadlines or exam dates to be altered for reasons of personal travel.
- Extension requests for reasons of illness MUST be accompanied by documentation from a medical professional stating the dates you were under their care and the date you were cleared to return to school/work.

Academic integrity

- You are expected to read and follow Purdue's guide to academic integrity.
- Behavior consistent with cheating, copying, and academic dishonesty is not tolerated. Depending on the severity, such behavior may result in a zero score on an assignment or exam, a failing grade for the class, or even expulsion. Purdue prohibits dishonesty in connection with any University activity. Cheating, plagiarism, or knowingly furnishing false information to the University are examples of dishonesty. (Part 5, Section III-B-2- a, University Regulations). Furthermore, the University Senate has stipulated that the commitment of acts of cheating, lying, and deceit in any of their diverse forms (such as the use of substitutes or taking examinations, the use of illegal cribs, plagiarism, and copying during examinations) is dishonest and must not be tolerated. Moreover, knowingly to aid and abet, directly or indirectly, other parties in committing dishonest acts is in itself dishonest. (University Senate Document 7218, December 15, 1972).
- A very detailed set of criteria that is enforced rigorously related to academic integrity is applicable to CS 17600. The consequences for violating course policies are serious both within the course and with the Office of the Dean of Students to determine whether you remain eligible to continue to study at Purdue University.
- You are encouraged to discuss any CS 17600 topic including high-level ideas about how to approach an assignment. But, under no circumstances will the exchange of, or shared access to, code via written or electronic means be permitted between teams for collaborative assignments or individuals for individual assignments.
- It is considered dishonest either to read another solution or to provide anyone with access to your work (or that of another student). Be mindful when working on code with others on individual assignments as this is discouraged. The work you submit must be your own original effort and not the result of unacceptable, even if unintentional, collaboration.
- **Minimum consequences for violating course policies will include:**
    - First offense, a zero for the assignment, a reduction of one letter grade at the end of the semester, AND a referral to the Office of the Dean of Students for disciplinary action.
    - Second offense, a zero for the assignment, a failing grade for the course, AND a referral to the Office of the Dean of Students for disciplinary action.
- **Notable exceptions to the first offense minimum consequences:**
    - Any violation on an exam will result in a failing grade for the course and a Dean of Students referral. Acts such as a misrepresentation of identity or location will result in a failing grade for the course and a Dean of Students referral.
    - Posting to, requesting from, or accessing solutions found on unapproved sources, particularly those found on-line, will result in a failing grade for the course and a Dean of Students referral.

Artificial intelligence (AI) policy

Students in this course are responsible for developing the programming, problem-solving, data wrangling, database, and analytical skills necessary to become effective data professionals. These skills require active practice and critical thinking that cannot be delegated entirely to AI tools. While AI/LLM technologies can generate code, explanations, and data analysis suggestions, they may also produce incorrect, inefficient, incomplete, or misleading results. Students are ultimately responsible for understanding, verifying, and being able to explain all submitted work.

Because students are expected to demonstrate their own understanding of course concepts and programming skills on quizzes, exams, and assignments, and because this course includes a live coding midterm exam in which students must independently write code without AI assistance, excessive reliance on AI tools may hinder learning and academic performance. **Therefore, students are encouraged to complete assignments primarily through their own efforts and use AI tools only as supplemental learning resources.**

Unless otherwise specified, students may use AI tools for learning support, debugging assistance, code explanations, brainstorming approaches, and improving the clarity of written work. **Students remain fully responsible for the accuracy, originality, and integrity of all submitted work.**

Nondiscrimination statement

Purdue University is committed to maintaining a community which recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her own potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life.

Accessibility

Purdue University strives to make learning experiences as accessible as possible. If you anticipate or experience physical or academic barriers based on disability, you are encouraged to contact the Disability Resource Center at: drc@purdue.edu or by phone: 765-494-1247. Attend instructor office hours to discuss any concerns regarding the implementation of your accommodations.

Emergency plan

In the event of a major campus emergency, course requirements, deadlines, and grading are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant information about changes in this course will be disseminated by the course email list and Purdue Brightspace system. Follow official university-issued instructions. Visit the Office of Emergency Preparedness website for more information.

Revision history

The course staff reserve the right to revise the syllabus and will provide notice of doing so.

- June 10: Initial version.